

Team 2 Research

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How hardware and software are used within the context of the industry

Software

- Software is utilized within Amazon's industry to make websites and apps that will expand their number of customers and to expand the number customers they are able to reach.
- Software is also used to increase the security of websites and users.
- Amazon's software has evolved to support their rapid growth by setting an industry standard for software.
- Software can also be used to collect data about the users. For example; malicious activity to increase security or to advertise apprenticeship, job or work experience offers.

Hardware

- Hardware is utilized within Amazon's industry to run the software made by developer's so that Amazon can advertise job offers and apprenticeships to expand the range of potential employees.
- Amazon's hardware is used to run their websites on a server for customers to access, this allows them to expand their business to more people that might want to apply for Amazon's job offers whereas without the hardware and software people probably won't not know about their company and this also applies to customers.
- Hardware is also used to increase the security of the server.

AWS (Amazon web services)

- Amazon stores their data on the cloud using AWS (See appendix A.1)
- Businesses can use AWS by paying amazon for some cloud storage space that will be protected and maintained by Amazon this means that other business don't need to pay for maintenance and don't need to run their own security like they would with a physical server.

How hardware and software could be used to expand Amazons reach of youth talent?

- Hardware and software could be used to connect to youth talent by creating a website or app that would allow students to view all the potential work experience and apprenticeships that are available. This

would increase the number of people that apply to the work experience and apprenticeships that Amazon offer.

- Having a website or app that people can access to look for work experience, job offers, and apprenticeships will make those offers be more accessible for people because the majority people now have access to the internet so they can look for work faster than in the past when people didn't have easy access to the internet

How can hardware and software be used to ensure security on Amazons website/apps?

- Hardware and software can be used to ensure security by using routers to prevent IP addresses, that are not authorised, gaining access to Amazons servers and causing harm to Amazon by leaking private data that Amazon stores.
- Hardware and software can be used to ensure security by using routers masking the server IP address as the routers IP address to hide the server IP from outside IP addresses this prevents IP addresses from pinging the server directly.
- Hardware and software can also be used to prevent DOS attacks from happening by having multiple servers on different sites so if one site were to go down then the other servers on different sites will be able to run until the other server is back up.
- Hardware and software can also be used by using air gapping to prevent DDOS attacks from being a risk as it physically separates the server from the internet.
- Software can be used to ensure security by using an anti-virus to prevent malware from being downloaded or injected onto a computer, which could access private information.
- Software can also prevent passwords from being leaked by hashing all the passwords that Amazon stores which encrypts them without being easily decryptable, this means passwords shouldn't be stolen even if a data breach were to happen.

Newly emerging technologies

AI:

ADVANTAGES: (Used different sources for the AI research)

Automatic, efficient responses

- AI responses are fast, allowing user's questions to be efficiently answered as they search through databases for specific answers that can relate to the inputted question.
- Responses will be 24/7, meaning that anyone could talk to the AI to get their answer without having to wait until customer service is available.
- Cost reduction, it will handle high volumes of inquiries, which automatically lowers labour costs and speeds up operations.

Enhanced Decision Making

- AI systems can analyse historical patterns and predict future trends, which can reduce human error and guesswork.
- AI has high speed processing, meaning that companies can react instantly to fast-changing market conditions as they can inquire about market conditions to the AI as well as choices for the future.
- AI surfaces hidden correlations in complex data, such as identifying early risk indicators in finance or medicine combinations in healthcare.

Reduction of Human Error

- AI has advanced pattern recognition, meaning that it will reduce human error as the AI will recognise correct answers as well as anomalies in inputted data, resulting in a change that will positively impact data.
- AI can predict analytics, this means that it can forecast potential system failures, which can lead to companies fixing systems that can reduce costs.
- It does not have fatigue as it is a system, meaning that it will focus on outputting data rather than getting distracted, tired or stressed like humans do.

Continuously Learning

- As AI is always available it is also able to continuously learn from human inputs as it can research sources and give out appropriate information to the user.
- Systems can easily adjust to changing environments, trends or shifting user behaviours, meaning that it can give information about new trends, leading to companies being able to advertise their products with a latest trend to increase profit.

DISADVANTAGES:

Job Replacement

- Easier jobs, like customer service is being replaced by AI, this leads to customers not having empathy from the company as AI lacks empathy, as well as jobs being taken away from customer service as they are all AI automated messages and responses.
- Replacing jobs in companies can often experience slower response times, higher error rates within the AI as well as reduced satisfaction from customers as the AI may not give correct results.

Bias and Discrimination

- It can have economic inequality which can limit job openings and career growth for specific genders or races by learning from biased past data.
- Companies can face heavy fines and lawsuits for breaking laws like the EU AI Act when the tools are used in a discriminatory way.
- People who face constant unfair automatic rejections or mistreatment experience higher levels of stress and anxiety.

Lack of Transparency

- AI systems can hide unfair rules or biased training data, this can lead to unfair treatment or discrimination against certain groups in areas like hiring, loans, or healthcare.
- When an AI makes a harmful or wrong choice, no one can figure out who is at fault as developers, companies and users can't trace back to the specific human action which trained the ai to respond in that way.
- Governments worldwide require clear rules for data privacy and ethical standards, hiding how an ai processes can break new laws and bring heavy fines.

High Cost

- Building AI tools require expensive hardware as well as specialised software, this can range for an investment with a value from thousands to millions of pounds.
- Systems need continuous updates, monitoring and retraining with data, leading to long-term operational budgets.
- AI requires a lot of energy to power the databases as well as the systems, which leads to high electricity consumption and high carbon emissions.

Has no Empathy or Creativity

- Empathy
 - AI can only mimic empathy through patterns and cannot feel any emotion, this can cause failures in sensitive fields such as mental health counselling or customer service.
 - Fake empathy from machines can trick users into emotional overreliance, this can lead to isolation or a breakdown in real human trust.
 - AI makes choices using cold responses, as it cannot weigh the moral and emotional toll the choices it makes that take on real people.
- Creativity
 - AI builds content by mixing past data, this means it cannot produce true original breakthroughs or unexpected artist visions.
 - Relying on AI flattens culture and design because the AI's system repeats the safer, common patterns rather than taking bold creative risks.
 - When an AI is faced with brand new challenges, AI struggles as it lacks comprehension and literal-only processing.

Overreliance

- **People rely on AI massively, which can lead to people accepting incorrect data that has been proven false and can change beliefs about a specific topic.**
- **It causes a loss of critical thinking as people rely on AI, which is increasingly dulling people's problem-solving skills and lowers the ability to spot misinformation.**

- **AI can 'hallucinate' facts, fabricate references or inherit historical human biases, which can lead to poor decisions if users accept the outputs without verification.**

Misinformation and Privacy Risks

- **Misinformation**
 - AI tools can easily generate realistic fake images, audio and videos that can trick viewers, this can manipulate public opinion as well as effect someone's reputation.
 - AI can often state incorrect facts with confidence; this can lead to users trusting wrong information and repeat false data.
 - Some people use AI to mass-produce fake news articles to make users believe it with using misinformation.
- **Privacy**
 - AI models need large amounts of data to learn, this means that they harvest text, photos and personal details from public websites that users have inputted without permission.
 - Companies might collect information for one specific task, for example, a job application, and then use it for training AI.
 - AI can analyse online habits to infer private details, such as health, beliefs, or financial status that were never meant to share.
 - People can do inversion attacks to reverse-engineer an AI model to extract private data which was used during the training phase.

Environmental Impacts

- It has a high energy demand, as powering servers that train and run the ai needs large amounts of electricity, this frequently comes from burning fossil fuels that release greenhouse gases, effecting the air in the atmosphere, resulting in climate change.
- It consumes large amounts of water as giant data centres generate extreme heat and needs gallons of water for cooling down the systems, this can strain local water suppliers.
- Mines resources, building the specialised computer chips requires mining finite minerals and metals, such as lithium, silicon and copper, this can degrade soil and disrupt local ecosystems.

- Rapid hardware upgrades and the discarded servers create a growing amount of electronic waste, which can contain hazardous materials that can pollute the environment.

Accessibility: (See Appendix B.1)

Auditory disabilities: (See Appendix B.2)

- Can range from Mild to Moderate, can have hearing loss in one or both ears.
- Some people with auditory disabilities can hear sounds but not sufficiently to understand all speech, especially when there's background noise.
- Usually rely on transcripts and captions of audio media, media players that have settings for captions, adjusting text size and colours of captions.
- High-quality foreground audio that is clearly separated from any background noise.
- Examples of Auditory disabilities:
 - Hard of hearing - Mild or moderate hearing impairments in one or both ears
 - Deafness - Substantial, uncorrectable impairment of hearing in both ears
 - Deaf-blindness - Substantial, uncorrectable hearing and visual impairments.
- Examples of barriers for people with auditory disabilities:
 - Audio content that does not contain captions or transcripts
 - Media players that do not contain captions or adjustable font settings.
 - Web-based services that rely on interaction using voice only.
 - Lack of sign language to supplement important information and text that is difficult to read.

Cognitive and Learning disabilities: (See Appendix B.3)

- Can affect how people store, retrieve or use information, often some functions are impaired while others are unaffected.
- Usually invisible and can be age-related, many may not have a formal diagnosis due to social stigma, vocational concerns and prejudice.
- Poor design or content choices can undermine or make impossible these different approaches.

- May affect the ability, to learn, communicate, read, write, do math, or process sensory input.
- May affect the ability of understanding or processing new or complex information or learn new skills.
- May affect the ability of using memory and attention or visual, language or numerical thinking.
- Examples of Cognitive and Learning Disabilities:
 - Age-related forgetfulness.
 - ADHD
 - Autism
 - Brain Injury
 - Dementia
 - Dyslexia
 - Dyscalculia
 - Learning disabilities
 - MCI
 - Mental Health disabilities
 - Multiple sclerosis
- Examples of barriers:
 - Complex, multistage processes such as forms
 - Complex or inconsistent navigation mechanisms and page layouts that are difficult to understand and use.
 - Complex sentences that are difficult to read and unusual words that are difficult to understand
 - Metaphors and other non-literal text whose meaning is not predictable from the usual meaning of words.
 - Long passages of text without images, graphs, or other illustrations to highlight the context.
 - Moving, blinking or flickering content and background audio that cannot be turned off.
 - Passwords and access codes that rely on memory.
 - Time-outs on activities.
 - Web browsers and media players that do not provide mechanisms to suppress animations and audio.
 - Visual page designs that cannot be adapted using web browser controls or custom style sheets.

Physical disabilities: (See Appendix B.4)

- Includes weakness and limitations of muscular control, limitations of sensations, joint disorders, pain that impedes movement, missing limbs.
- People that use specialised hardware and software such as:
 - Ergonomic or specially designed keyboard or mouse.
 - Head pointer, mouth stick, other aids to help with typing.
 - On screen keyboard with trackball, joysticks or other pointing devices.
 - Switches operated by foot, shoulder, sip-and-puff, and other movements.
 - Speech recognition, eye tracking, other approaches for hands-free interaction.
- Examples of physical disabilities:
 - Amputation
 - Arthritis
 - Fibromyalgia
 - Rheumatism
 - Reduces dexterity
 - Multiple sclerosis
 - Muscular dystrophy
 - RSI
 - Tremors and spasms
 - Quadriplegia
 - Seizure disorders
- Examples of barriers for people with physical disabilities:
 - Websites, web browsers and authoring tools that do not provide full keyboard support.
 - Insufficient time limits to respond or to complete tasks, such as to fill out forms.
 - Controls, including links with images of text, that do not have equivalent text alternatives.
 - Missing visual and non-visual orientation cues, page structure and other navigational aids.
 - Inconsistent, unpredictable and overly complicated navigation mechanisms and page functions.

Speech disabilities: (See Appendix B.5)

- Includes difficulty producing speech that is recognisable by others or by speech recognition software.

- Can struggle with communication to automated hotlines, applications or other software that may be operated by using voice commands.
- Examples of speech disabilities:
 - AOS
 - Cluttering
 - Dysarthria
 - Speech sound disorder
 - Stuttering
 - Muteness
- Examples of barriers for speech disabilities:
 - Web based services, including web applications that rely on interaction using voice only.
 - Websites that offer phone numbers as the only way to communicate with the organisations.

Visual disabilities: (See Appendix B.6)

- Can range from moderate to mild, vision loss in one or both eyes (low vision) to substantial vision loss in both eyes (blindness)
- Some can have reduced sensitivity to certain colours (colour blindness) or increased sensitivity to bright colours.
- They can rely on enlarging or reducing text sizes and images, customising settings for fonts, colours and spacing, listening to text to speech synthesis of the content, listening to audio descriptions of video in multimedia, reading text using refreshable braille.
- Developers should use style sheets to separate content from its presentation and correctly code the structure so it can be processed and presented in different ways by browsers and assistive technologies.
- Some people only see small portions of content at a time or are perceiving the colours and design differently. People use customised fonts, colours, and spacing to make the content more readable or are navigating through the content using keyboard only because they cannot see the mouse pointer.
- Example of visual disabilities:
 - Colour vision deficiency
 - Low vision
 - Blindness
 - Deaf blindness

- Examples of barriers with visual disabilities:
 - Images, controls and other structural elements that do not have equivalent text alternatives.
 - Text, images and page layouts that cannot be resize or that lose information when resized.
 - Missing visual and non-visual orientation cues, page structure, and other navigational aids.
 - Video content that does not have text or audio alternatives, or an audio-description track
 - Inconsistent, unpredictable and overly complicated navigation mechanisms and page functions.
 - Text and images with insufficient contrast between foreground and background colour combinations.
 - Websites, web browsers and authoring tools that do not support the use of custom colour combinations or do not provide full keyboard support.

How digital solutions could be used to meet different user needs

Information about digital solutions

- Digital solutions are designed to provide a personalized and accessible experience for users with disabilities to be able to access and interact with technology without any difficulties. Having accessibility features such as screen readers and text-to-speech would ensure that digital solutions are inclusive and usable for users to navigate through the internet. This type of solution is used to improve user experiences which can increase users' satisfaction as it emphasizes on designing digital products for user's needs which can be used effectively against digital divide. (See Appendix C.1)
- Digital solution is a technological tool that's used to solve problems and improve efficiency for a business. For example, UCD (User-centred design) focuses on users by using product design and development to match with user needs. This tool has been integrated into many places from economics to businesses. (See Appendix C.2)
- In the business sector, digital solutions are tailored to meet business' needs and requirements by helping with automated tasks and processes which

allows businesses to deal with manual work. Roughly, automated solutions reduce 60% of manual work. Cloud services and software applications are commonly used in the industry. With variations of different types of digital solutions, around 40% of operational efficiency is improved and reduces operational costs. With the implementation of digital solutions, around 50% of digital information initiatives meet the business targets and over 90% of cloud technologies are used in organizations. Additionally, digital solutions are centred around businesses with significant investment and progress. They continue to grow significantly and contribute to the economy, highlighting how digital solutions hold importance to innovation. (See Appendix C.3)

Advantages and disadvantages

- One of the advantages of using digital solutions is how cost-effective it can be. Implementing digital solutions can reduce operational costs and eliminate the need for physical infrastructure which saves enough time to maintain digital services. Another advantage of using digital solutions is customization. This part of the process of creating digital solutions can be tailored to adapting to users' needs and preferences by having a wide range of options for the users to select. This can drive innovation and improve service quality by applying continuous updates to the digital products and listening to users' feedback. From having feedback, businesses can continue to develop more options for users to make it easily accessible to everyone. (See Appendix C.4)
- However, one of the disadvantages is the security risks. While digital solutions offer a lot of flexibility and benefits to businesses, digital technology is at risk of exposing sensitive information, leading to data breach and loss of privacy. Another disadvantage is the barrier to digital services. Some users may struggle to transition to digital services from the lack of access to technology or understand how to use it. A digital divide prevents some people from having access to the internet or having any affordability to buy and access one. This can lead to the users at a digital disadvantage as they're being excluded by the digital barrier. (See Appendix C.5)

How can digital solutions be used to meet different user's needs on Amazon?

- Digital solutions could be used to meet user's needs by providing sources that heavily focus on customer service and customer satisfaction. For example, data personalization is added to Amazon which would be tailored to product

recommendations and marketing, ensuring that the users have highly personalised shopping experience.

- Making a user-friendly interface to the users would make the shopping experience simpler and more intuitive, increasing customer's satisfaction and adaptation to the website.
- Amazon would ensure that users would have a responsive customer service by offering 24/7 customer support and a return policy, this helps to maintain high levels of customer satisfaction and meet the customers' standards.

The industry-specific guidelines and regulations you will need to follow

Copyright

Copyright is a crucial aspect of regulation when developing a website for a company. For example, copying code straight from other websites without permission and copying ideas is plagiarism. To avoid this, it's important to keep designs unique to not dispute legal action. We can use open-source repositories such as GitHub to access a library code that can be utilised efficiently to use samples of code and the can be personalised to the set brief. Imagery used must be taken from open sources to not infringe upon laws also. This can be performed by visiting websites that allow images to be used for free. This is a cost-effective solution as we don't have to pay licenses for certain imagery as working with amazon, we should have access to Intellectual property such as logo's we can utilise so the branding can be professionally displayed. The brief has linked websites that cover the use of trademarks and the use of Amazon's branding.

Data Protection

The GDPR (General Data Protection Regulation), effective since May of 2018, is a protection act that companies must follow legally. The GDPR defines how companies may process client data and sets rules around their rights to privacy and how companies use, store and transfer data without breaching the clients forementioned rights. This is why it is important to have a secure database where we can safely store personal

information the user will enter. This will prevent legal actions from the company and keeping good reputation with clients.

Cultural appropriation

Throughout the project, we will take proactive measures to prevent cultural appropriation and mitigate algorithmic bias. AI frequently inherits bias in training data which leads to stereotypical portrayals. Our team doesn't rely purely on automation and must implement human action also. Another important factor are fonts and using novelty fonts that mimic ethnic backgrounds that shouldn't be included as it creates an unethical layout that may offend people.

Appendix A

A.1:[What is Data Store? - Data Store Explained - AWS](#)

A.2:[DoS Attacks: Types and Defense Strategies](#)

Appendix B

B.1:[Diverse Abilities and Barriers](#)

B.2:[Auditory](#)

B.3:[Cognitive and learning](#)

B.4:[Physical](#)

B.5:[Speech](#)

B.6:[Visual](#)

Appendix C

C.1: [30 Pros And Cons Of Digital Technology](#)

C.2: [User-centred design consultancy and services](#)

C.3: [Top 100 Digital Transformation Statistics \[2026\]](#)

C.4: [Creating digital Solutions That Resonate with Real-World Needs](#)

C.5:[Digital adoption and digital disadvantage today: what has changed, and what barriers remain?](#)

